KernelScript: Compiler-Checked Cross-Boundary Typed DSL for eBPF Applications

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Abstract

eBPF has become the standard mechanism for extending Linux with custom packet processing, tracing, and security policies, with a verifier check before execution to guarantee it will not crash the kernel. The programming model, however, is fragmented. A single application spans kernel code, a userspace loader, and shared maps, yet the relationships among them (types, lifecycle ordering, shared state) go unchecked. An event struct defined differently on each side silently garbles every record, and a map type mismatch silently corrupts shared state. We observe that these cross-boundary relationships are duplicated concepts that a type system can unify. We present KernelScript, a DSL that types maps, program handles, and execution domains in one source, then lowers to standard C through the clang/bpftool/libbpf toolchain. The compiler detects mismatches, such as wrong context types or map type disagreements, before code generation. Generated projects load, attach, and pass traffic on a real kernel, showing that eBPF's cross-boundary structure can be typed while preserving the existing toolchain.

CCS Concepts: • **Software and its engineering → Domain specific languages**; *Static analysis*; • **Computer systems organization** → *Reliability*.

Keywords: eBPF, domain-specific language, static typing, systems

1 Introduction

eBPF lets developers run application-specific logic inside the Linux kernel under a verifier that proves memory safety and bounded execution [21]. It now underpins packet processing in major cloud providers, powers observability platforms, enforces security policies, and enables scheduler experimentation within a large and growing ecosystem [23].

Yet the programming model is fragmented. A single eBPF application spans kernel code constrained by the verifier, a userspace loader that opens objects and manages lifecycle, shared maps and ring buffers, and (for newer features) generated skeleton headers that must match the running kernel. The relationships among these pieces, such as a map's key and value types or the order of load, attach, and detach, are checked only at load time or not at all. Errors may go entirely undetected because they do not break the kernel. When a map's value struct is redefined in kernel C but not in the userspace loader, data is silently misinterpreted at runtime. When a skeleton header becomes stale after a struct change, userspace compiles against the old layout and reads corrupt fields.

Existing tools reduce boilerplate yet leave cross-boundary relationships implicit. The standard eBPF development workflow requires hand-written kernel C, userspace C, and a Makefile [15]. Aya compiles both sides from Rust and shares typed maps [1], and cilium/ebpf does similarly in Go [3]. In both, lifecycle ordering remains a runtime property. bpftrace targets tracing with higher-level abstractions [2] rather than general eBPF applications. No existing tool types the full cross-boundary structure at compile time.

The key observation is that these relationships are single concepts duplicated across components. A map declaration is simultaneously a kernel data structure, a userspace lookup target, and a type contract. A program handle is simultaneously a function name, a BPF section, and a file descriptor. In C/libbpf, naming conventions connect these copies, so mismatches surface late. In a unified typed source, the compiler can see and check them directly.

KernelScript realizes this idea as a compiled, multi-domain DSL. A developer writes eBPF entry points, userspace control flow, maps, and ring buffers in one source file. The compiler types maps, program handles, and execution domains, then lowers to eBPF C, userspace C, and a Makefile. The standard libbpf and bpftool path then produces the final binaries, preserving the existing toolchain.

This paper contributes:

- A *typed model of cross-boundary eBPF structure*: execution domains, typed maps, and program-lifecycle handles, each with a typing rule and matching compiler diagnostic (Section 3).
- A compiler that realizes this model while preserving the stock kernel toolchain, so compatibility failures remain visible (Section 4).
- An evaluation showing that each invariant triggers a diagnostic, that one source generates buildable libbpf projects, and that generated objects load, attach, and run on a real kernel (Section 5).

2 Background and Motivation

eBPF allows user-supplied bytecode to run inside the Linux kernel after a verifier proves memory safety and bounded execution [21, 23]. The mechanism now underpins a wide range of applications: tracing and profiling [2, 12], high-performance networking [9], security policy

enforcement [14], CPU scheduling [11], page-cache customization [30], GPU resource control [28], and userspace extension runtimes [27]. Every eBPF application has a kernel program and a userspace loader that communicate through maps or ring buffers. The standard eBPF development workflow requires developers to maintain separate kernel C, userspace C, and a Makefile [15]. Aya unifies both sides in Rust [1], and Rex replaces the verifier with language-level safety in Rust [13], but neither types the full cross-boundary structure at compile time.

The kernel/userspace split introduces three kinds of coupling that current tools do not fully address. *Lifecycle coupling* requires operations to occur in a valid order (load before attach, detach before close), but C/libbpf checks this only at runtime. *Representation coupling* requires a map's key and value types to agree across kernel code, generated headers, and userspace lookups, yet a mismatch is silent until something fails. *Compatibility coupling* arises because generated bindings depend on the exact kernel and libbpf versions. The verifier guarantees memory safety but not application correctness across the boundary: calling `attach` before populating maps lets the kernel program run against empty state. Redefining a ring-buffer event struct on one side without updating the other silently garbles every record. A stale skeleton header after a kernel struct change compiles without warning but misaligns every shared field. Automated eBPF-to-Rust migration confirms that these implicit cross-boundary relationships are a primary source of translation errors [4]. A DSL can move lifecycle and representation checks to compile time.

3 A Typed Model of Cross-Boundary Structure

Every rule corresponds to a diagnostic exercised by the rejection suite in Section 5.

A KernelScript program is a flat sequence of declarations. Listing 1 shows the minimal shape of an application. The rest of this section makes its three relationships precise.

Listing 1. A unified-source eBPF application.

```
include "xdp.kh"

@xdp fn pass_all(ctx: *xdp_md) -> xdp_action {
  return XDP_PASS
}

fn main() -> i32 {
  var prog = load(pass_all)
  attach(prog, "lo", 0)
  detach(prog)
  return 0
}
```

To address representation coupling, each function carries an attribute that assigns it to an execution domain $d \in \{user, ebpf, helper, kfunc\}$. The attributes `@xdp`, `@tc`, `@probe`, `@tracepoint`, and `@perf_event` mark eBPF entry points, while `@helper` marks kernel-shared eBPF code and `@kfunc` marks kernel-module code. An unattributed `fn` is userspace. The checker tags each function with its domain and rejects illegal crossings, such as a userspace function calling an `@helper`. Each eBPF attribute also fixes a context and return type: for example, `@xdp` requires `ctx:*xdp_md` returning `xdp_action`, and `@tc` requires `*__sk_buff` returning `i32`. The compiler rejects any signature that disagrees with its attribute before code generation. The same surface syntax thus remains domain-aware: userspace can allocate and print freely, while eBPF code must obey stack and helper restrictions. struct_ops callbacks fit the same scheme, with context and return types determined by the registered slot.

To address representation coupling for shared state, a map is declared as a typed global (`var m : hash<K,V>(N)`) rather than a C section plus separate userspace lookup code. The compiler checks element types at every access site on both sides of the kernel boundary: an index `m[k]` requires `k:K` and yields `V`, in eBPF and userspace alike. A wrong key type, wrong value type, or access to an undeclared map is a compile-time error. One declaration thus replaces kernel definitions, generated fields, and userspace lookups that otherwise must agree by hand.

To address lifecycle coupling, the type system distinguishes a `FunctionRef` (a reference to an `@`-attributed function) from a `ProgramHandle` (a loaded program). The typing ensures that only `load` produces a `ProgramHandle`. The formation rule for `FunctionRef` connects lifecycle to domains: only an eBPF entry-point function inhabits `FunctionRef`, so an `@helper`, `@kfunc`, or userspace function cannot be passed to `load`.

$$\frac{f \text{ has an eBPF entry attribute}}{\Gamma \vdash f : \texttt{FunctionRef}}$$

$$\frac{\Gamma \vdash f : \texttt{FunctionRef}}{\Gamma \vdash \texttt{load}(f) : \texttt{ProgramHandle}}$$

$$\frac{\Gamma \vdash h : \texttt{ProgramHandle}}{\Gamma \vdash \texttt{detach}(h) : \texttt{unit}}$$

$$\frac{\Gamma \vdash h : \texttt{ProgramHandle}}{\Gamma \vdash \texttt{attach}(h, t, fl) : \texttt{u32}}$$

Because `attach` and `detach` require a `ProgramHandle` and only `load` produces one, "attach before load" is a *type* error, not a runtime failure. Passing a bare `FunctionRef`, an integer, or a string to a lifecycle builtin is a type error. The typing rules enforce a producer/consumer handle discipline. Because only `load` produces a `ProgramHandle`, the system enforces the single ordering constraint $load \prec \{attach, detach\}$ rather than the complete load-to-attach-to-detach protocol. It guarantees that lifecycle builtins receive values of the right kind. Full path-sensitive properties such as use-after-detach,

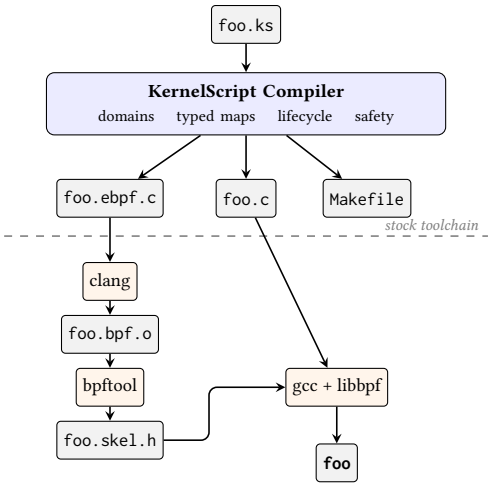

Figure 1. Compilation pipeline. The compiler checks cross-boundary invariants and emits C; the stock clang/bpftool/libbpf toolchain produces the final binary.

double-attach, or leaked handles remain outside the type system's scope.

The benefit is localized compiler errors earlier in development, replacing naming conventions and late verifier or attach-time failures. Section 5 measures whether these diagnostics trigger. Section 6 explains the limits of that evidence.

4 Design and Implementation

The compiler has three design goals. First, it must detect cross-boundary errors at compile time, before the verifier or runtime sees them. Second, it must preserve the existing Linux BPF toolchain so that type information, portability, and verifier diagnostics remain available. Third, it must generate idiomatic C that developers can inspect and debug.

The type system draws on established ideas. Typestate tracks object state in types [20], and linear types enforce single-use resource protocols [5, 10, 24]. KernelScript applies a lightweight form of these ideas, distinguishing producer and consumer handles and annotating execution domains without full substructural typing.

To meet these goals, the compiler (written in OCaml with dune) parses KernelScript source, resolves imports and includes, builds symbol tables, runs multi-program analysis and type checking, performs safety analysis, builds an IR, and emits C (Figure 1). The compiler comprises nine modules covering maps, codegen, safety, and struct_ops. For an input `foo.ks`, the generated project contains `foo.ebpf.c`, `foo.c`, a `Makefile`, and, when kfuncs are present, `foo.mod.c`. The Makefile invokes bpftool to dump `vmlinux.h`, clang to produce a BPF object, bpftool to generate a skeleton, and gcc to link the userspace loader against libbpf (together with libelf and zlib).

Each typing rule of Section 3 maps to a concrete compiler pass. The compiler attaches a domain to each function as a scope refined by the entry attribute. The IR records each function's program type so that code generation can dispatch the same source function to the eBPF or userspace backend. Typed maps lower to a single descriptor that records key and value types once. Both backends read that descriptor to emit the eBPF `SEC(".maps")` definition and the userspace accessor (Listing 2). Lifecycle handles are enforced in the front end. Only `load` yields a `ProgramHandle`, and `attach` and `detach` require one, so a misordered call fails type checking before code generation. The handle then lowers to a file descriptor via libbpf APIs.

Listing 2. One typed map declaration lowers to both a kernel `SEC(".maps")` definition and a userspace accessor from a single descriptor.

```
// KernelScript source
var counts : hash<u32, u64>(1024)

// generated eBPF C
struct {
  __uint(type, BPF_MAP_TYPE_HASH);
  __uint(max_entries, 1024);
  __type(key, __u32);
  __type(value, __u64);
} counts SEC(".maps");

// generated userspace C: same key/value types
__u32 key = ...; __u64 val;
bpf_map_lookup_elem(counts_fd, &key, &val);
```

The compiler does not replace the verifier, clang, bpftool, or libbpf. Existing diagnostics therefore remain available, and any compatibility failure stays visible. For interfaces that move faster than the DSL can stabilize, `include` and `extern` declarations provide a fallback to raw header declarations.

5 Evaluation

The evaluation answers three questions. *(Q1)* Does each typed cross-boundary invariant trigger a compile-time diagnostic? *(Q2)* Does a single source generate a complete, buildable libbpf project, and does unifying it reduce maintenance surface? *(Q3)* Does generated code load, attach, and run on a real kernel? All experiments run on Linux 6.15 with clang 18, bpftool 7.7, and libbpf 1.3.0. SLOC counts exclude blank lines, comments, and generated headers.

5.1 Q1: Compile-Time Diagnostic Coverage

The compiler test suite contains 85 suites and 1095 tests with no failures. These tests cover all major subsystems from lexer to code generation. A separate rejection suite of 28 targeted programs exercises the diagnostics of Section 3 directly, with one positive control and 27 expected rejections covering lifecycle misuses, program-signature violations, map type errors, domain crossings, and safety violations. Each case carries an expected diagnostic substring, so incidental rejections do not count as passing.

To measure whether KernelScript detects mistakes *earlier* than the stock toolchain, we inject four realistic mistakes into a working XDP map-counter in both KernelScript and matched hand-written C/eBPF, recording the earliest rejection stage along a shared ladder (compile < build < verifier < attach < runtime < undetected). As Table 1 shows, KernelScript rejects all four at compile time, while C/eBPF ties on all but the cross-boundary case. That case passes an `@xdp` entry point a `*__sk_buff` context instead of `*xdp_md`. KernelScript rejects it via the program-signature rule of Section 3, whereas the C version compiles and loads silently because the body never dereferences the context, giving the verifier no invalid access to flag. A body that did read a context field could trip the verifier, though it would report the access violation rather than the underlying context-type confusion. The other three mistakes (undeclared map, wrong value type, oversized stack) tie because clang catches them locally. Typing therefore helps precisely on cross-boundary relations that local C checking cannot see. Section 6 discusses the limits of this author-constructed set.

Table 1. Earliest detection stage per injected mistake (BS = KernelScript). KernelScript is strictly earlier only on the cross-boundary context-type mistake; the rest are ties clang already catches.

Injected mistake	Coupling	BS	C/libbpf
Wrong context type	signature/domain	compile reject	undetected
Undeclared map	representation	compile reject	compile reject
String into `u64` value	representation	compile reject	compile reject
Oversized stack buffer	safety	compile reject	compile reject

5.2 Q2: Project Generation

Of 44 example programs, 43 compile successfully and 41 produce working libbpf projects. The single compile-time rejection, `safety_demo.ks`, demonstrates early error detection. Safety analysis flags 608 bytes of stack use (above the 512-byte eBPF limit) and halts before codegen. The two struct_ops programs that compile but fail to build expose toolchain version skew. Their bpftool-generated skeletons reference a libbpf field absent from the host headers (Section 5.3 confirms this is not an inherent limitation). For successfully built examples, the compiler emits a median of 472 lines from 31 source lines. Across 11 workloads with hand-written C/libbpf baselines, KernelScript sources total 203 lines versus 1105. The examples exercise the major eBPF features: XDP, TC, kprobe, tracepoint, perf_event, maps, ring buffers, tail calls, kfuncs, and struct_ops.

Hand-porting three `xdp-tutorial` [22] XDP programs to KernelScript yields comparable line counts to the original C/eBPF. A source-only scan of three public eBPF repositories confirms that the program types and map patterns we generate (XDP, TC, kprobe, ring buffers, struct_ops) appear together in real projects.

Source-footprint totals measure the size of a finished workload but miss how many places a requirement change must touch. To measure this change-amplification coupling, we construct five matched micro-edits (changing a map from `array` to `percpu_array`, changing a program from `@xdp` to `@tc`, moving a `perf_event` attach from standalone to grouped, extending a shared ringbuf event schema, and adding ringbuf reporting with userspace handling) and compare diff sizes between KernelScript and hand-written C/libbpf. Table 2 reports file count, edit-site count, changed LOC, and required manual synchronization across kernel code, userspace code, or skeleton/section conventions.

Table 2. Change amplification for five matched micro-edits (BS = KernelScript, C = hand-written C/libbpf). "Sync" reports which manually synchronized surfaces change: kernel (K), userspace (U), or skeleton/section conventions (S).

Edit	Impl.	Files	Sites	LOC	Sync
Map `array`→`percpu_array`	BS	1	1	2	—
	C	2	7	19	K/U/S
Program `@xdp`→`@tc`	BS	1	2	8	—
	C	2	6	30	K/U/S
Perf event standalone→grouped	BS	1	1	6	—
	C	1	8	91	U
Shared event schema	BS	1	3	4	—
	C	2	4	7	K/U
Add ringbuf reporting/handling	BS	1	5	23	—
	C	2	11	89	K/U/S

None of the KernelScript edits require manual cross-boundary synchronization, whereas hand-written C/libbpf requires userspace synchronization in 5/5 cases, kernel-side synchronization in 4/5, and skeleton or section-convention synchronization in 3/5. Diff-size medians reinforce this: KernelScript touches 1 file, 2 edit sites, and 6 lines, versus 2, 7, and 30 for C/libbpf. With only five edits these numbers are illustrative, but they show that a unified typed source reduces change amplification beyond total line count. As cross-boundary structure accumulates (pass-only XDP → counter map → ring-buffer reporting), KernelScript grows more slowly: +2 lines for the counter map versus +11 in C/libbpf, and +12 for ring-buffer reporting versus +202. By the reporting stage, C/libbpf reaches 9× the KernelScript source size.

5.3 Q3: Runtime Validation

Table 3 summarizes a progression of increasingly stringent checks on a real kernel: build success, verifier acceptance, attachability, and runtime behavior.

Table 3. Real-kernel checks: compile, build, verifier-load, and attach.

Stage	Population	Pass
KernelScript compile	44	43
Generated project build	44	41
Verifier-load (strict)	41	37
Isolated XDP attach/detach	27	27

Of 41 fully built objects, 37 pass the verifier. The four failures are a reference-ownership leak, a map-creation failure, a missing kfunc symbol, and an object with no program section. For XDP, all 27 eligible objects attach and detach successfully. Additional workloads exercise TC and struct_ops.

Matched workloads confirm the generated objects run correctly. Under `BPF_PROG_TEST_RUN`, a minimal generated XDP pass program matches its hand-written baseline in instruction count and latency. Over ten veth/iperf3 trials, both generated and hand-written XDP and TC programs deliver comparable throughput. These numbers are sanity checks for functional equivalence, not performance claims.

Preserving the kernel toolchain means KernelScript inherits its compatibility coupling. The two struct_ops build failures are not parser or type errors. Both compile to valid eBPF objects but fail because the bpftool-generated skeleton references a libbpf field absent from the installed headers. A direct libbpf runner bypassing skeletons loads both the generated and a matched hand-written tcp-congestion struct_ops object successfully. This confirms toolchain version skew rather than an inherent limitation. This is exactly the compatibility coupling of Section 2 that source unification cannot eliminate.

6 Limitations

The differential probe injects four hand-chosen mistakes rather than bugs mined from revision histories. Lifecycle typing enforces a producer/consumer handle distinction without full typestate. Use-after-detach and leaked handles therefore remain possible. The example programs are author-written. The external port shows that line-count savings vary when porting existing code rather than writing from scratch. The change-amplification study uses small checked-in micro-edits instead of real revision histories. It measures maintenance-surface coupling, not actual developer time or debugging effort. Traffic numbers are local sanity checks, not performance claims. Finally, the struct_ops build failures reflect bpftool/libbpf version skew, not a deep limitation. They do show that generated builds depend on toolchain alignment.

7 Related Work

Prior work on eBPF development falls into three categories. Domain-specific DSLs target particular use cases: bpftrace for tracing [2], BPFBox and ActPlane for security policies [7, 26], BMC for caching [8], P4c-eBPF for networking [17], and Yaksha-Prashna for bytecode analysis [19]. These DSLs primarily target kernel-side code and do not type the full cross-boundary structure that KernelScript addresses. General eBPF development tools include the standard eBPF development workflow [15], Aya for Rust [1], Rex for safe Rust kernel extensions without a verifier [13], Wasm-bpf for WebAssembly deployment [29], and eunomia-bpf for dynamic loading [6]. These reduce boilerplate but leave lifecycle and representation relationships implicit. Verified approaches take a different direction: BeePL provides correct-by-compilation kernel extensions [18], and Jitk and Serval verify JIT correctness [16, 25]. KernelScript occupies a middle ground. It generates both kernel and userspace code from one source and types cross-boundary relationships. The goal is practical integration with the existing toolchain, not formal verification.

8 Conclusion

KernelScript treats the cross-boundary structure of an eBPF application (program lifecycle, shared maps, and execution domains) as something a compiler can type, replacing conventions and late verifier/attach-time discovery. The evaluation confirms that each typed invariant triggers a compile-time diagnostic, that a single source generates buildable libbpf projects, and that generated objects load, attach, and pass traffic on a real kernel. The result is a concrete design point: typing eBPF's cross-boundary structure is feasible. Because the compiler lowers through the stock toolchain, the type system's guarantees and the version-skew limits it cannot absorb both remain visible and measurable.

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